

Katihar Engineering College, Katihar
3rd Semester (CSE) – Data Structures Laboratory

Programming Test – I

Duration: 75 Minutes

Maximum Marks: 15

General Instructions

1. Write, compile, and execute any **three** programs.
2. Use C as prescribed for the laboratory.
3. Appropriate data structures must be used for each problem.
4. Demonstrate the output using the test data given in the question.
5. Marks will be awarded for logic, implementation, testing, and appropriate handling of boundary cases.

Q1. Inventory Record Management

[5 Marks]

A small shop stores product quantities in an array:

25 40 15 60 35 50

Write a program that allows the shopkeeper to:

- Add a new quantity at a specified location.
- Remove a quantity from a specified location.
- Find whether a particular quantity exists.
- Display all current quantities.

Perform the following operations:

Insert 45 at position 3

Delete the element at position 5

Search for 50

Display the final array

Your program should also handle an invalid position appropriately.

Q2. Efficient Sequence Computation**[5 Marks]**

A scientific application needs the following sequence:

1 1 2 3 5 8 13 21 ...

The program must generate the first N terms.

Implement two versions:

- (a) Calculate a term using recursion.
- (b) Maintain previously calculated terms in an array and reuse them.

Use $N = 20$.

Your answer should include:

- i. Output from both approaches.
- ii. Explanation of why one approach performs substantially more repeated calculations.
- iii. Time and space complexity of both approaches.
- iv. Explanation of the memory-versus-computation trade-off.

Q3. Web Browser Navigation**[5 Marks]**

A web browser needs to maintain the pages visited by a user so that pressing **Back** returns to the most recently visited page.

The user visits:

google.com
university.edu
library.org
news.com

The user then presses **Back** twice.

Design and implement the required data structure to simulate this behavior.

Display the page reached after each Back operation.

Q4. HTML Tag Validator**[5 Marks]**

A simplified HTML editor needs to check whether opening and closing symbols are properly nested.

For the following strings:

< [()] >

and

< [(]) >

Write a program to determine whether the symbols are correctly balanced.

Treat the following as matching pairs:

$\langle, \rangle$ $[,]$ $(,)$

Q5. Expression Translator**[5 Marks]**

An embedded calculator receives the following expression:

$A+B*(C-D)$

Before evaluation, the expression must be converted into a representation in which each operator occurs **after the operands on which it operates**.

Write a program to perform this conversion.

Your program must correctly handle:

+ - * / ^
()

— End of Programming Test I —